COMPUTATIONAL PSYCHOLINGUISTICS · MID-EVALUATION

Lossy-Context Surprisal for Predicting Eye Movement in Reading

Do forgetting-based language models explain when and why readers look back?

Lakshmi Prajna Penmetsa · Sreevidhu Rithwik Yarra

Motivation

Standard surprisal assumes perfect memory — but human memory decays over context.

Eye-tracking captures multiple stages (FFD → TRT) but most models treat them uniformly.

Do early vs. late reading measures reflect different memory windows?

KEY INSIGHT

Futrell et al. (2020) proposed lossy-context surprisal:

weight context by β^distance

β = 1.0	Standard surprisal (full context)
β → 0	Local context only (strong decay)

Validated on self-paced reading only.
Never tested on eye-tracking — until now.

Research Questions

RQ1

Does lossy-context surprisal explain variance in eye-tracking measures beyond standard surprisal and dependency-length predictors?

RQ2

Does the optimal memory decay parameter β differ across early measures (FFD, GD) vs. late measures (GPT, TRT, regressions)?

→ Consistent with distinct stages of lexical vs. integrative processing

Methods: Analysis Pipeline

ZuCo-1 SR

12 subjects
354 sentences
Eyelink II

ET Extraction

FFD, GD, GPT
TRT, nFix, reg
Pixel bounding boxes

GPT-2 Surprisal

Subword alignment
Word-level bits
600 sentences

Lossy-Context Surprisal

$\beta \in \{0.1–0.9\}$
50 samples/word
Futrell et al. 2020

glmmTMB Models

Gamma & binomial
Random effects
AIC comparison

COVARIATES IN ALL MODELS

- Word Length
- Log Frequency (Zipf)
- Word Position
- Dependency Length (spaCy)
- Standard Surprisal (GPT-2)

Dataset & Eye-Tracking Measures

12	354	74,163	46%	500 Hz
subjects	sentences	word tokens	fixation coverage	eye-tracker

FFD	EARLY	Duration of 1st fixation on word — lexical access speed
GD	EARLY	Sum of first-pass fixations — initial processing effort
GPT	LATE	All fixations until rightward exit (incl. regressions)
TRT	LATE	Sum of all fixations on word — total processing
reg	LATE	Binary: any fixation coming from the right?

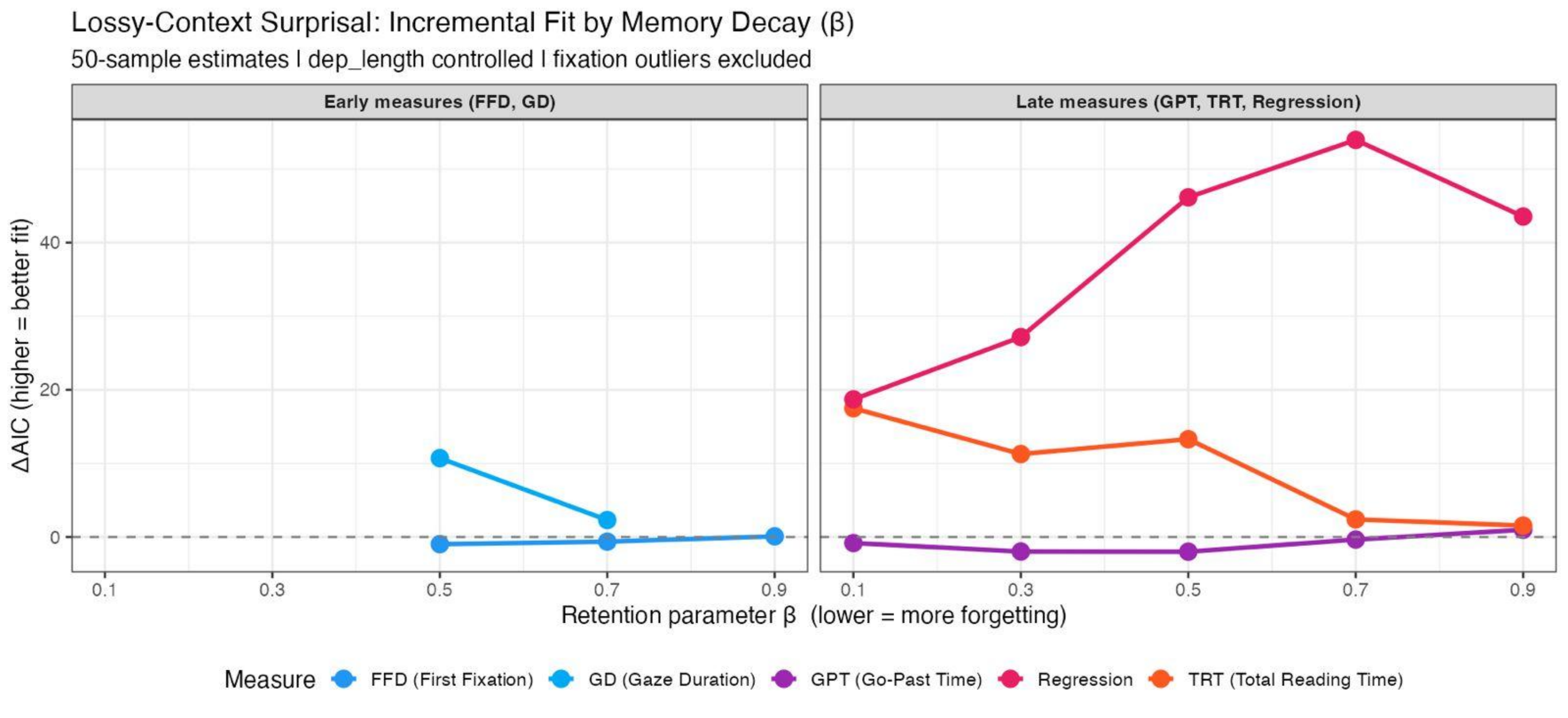
Baseline Model Results

Standard surprisal + word length + frequency + position + dependency length → glmmTMB

Measure	Surprisal β	p-value	Dep. Length β	p-value	Interpretation
FFD	−0.0074	0.024 *	+0.0043	0.056	Surprisal slows initial fixation
GD	+0.0012	0.792	+0.0049	0.104	No effect at first-pass stage
GPT	−0.0084	0.177	+0.0134	0.002 **	Dep. length drives regression-inclusive reading
TRT	+0.0028	0.612	+0.0048	0.209	Neither significant for total re-reading
reg	+0.0928	< 0.001 ***	+0.0397	0.006 **	Both predict regressions strongly

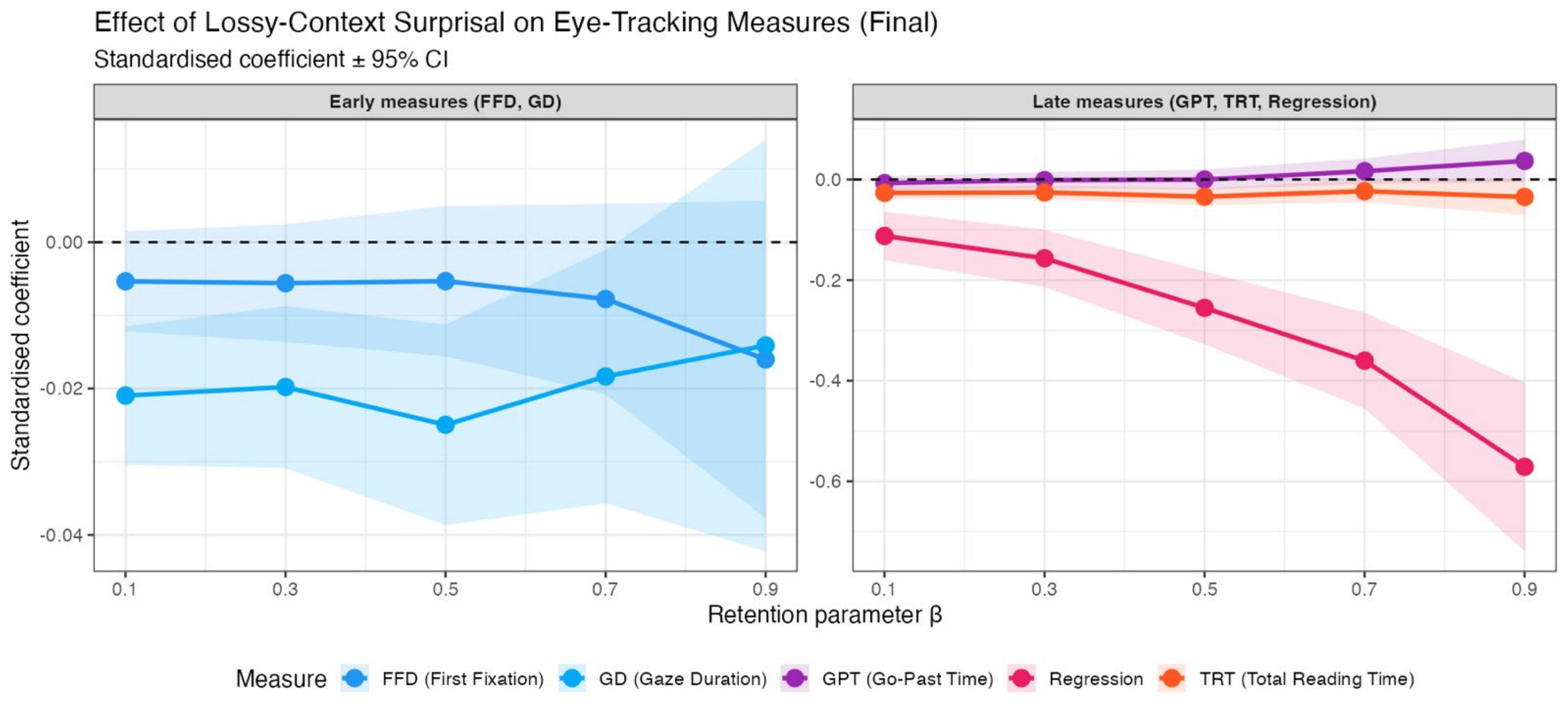
Lossy Surprisal: Fit by Memory Decay Parameter β

50-sample estimates | dep_length controlled | fixation outliers excluded



Effect of Lossy-Context Surprisal on Eye-Tracking Measures

Standardised coefficient ± 95% CI | Early measures (FFD, GD) vs. Late measures (GPT, TRT, Regression)



Key Findings

FFD

No Effect

Lossy surprisal does not predict first fixation — lexical access is memory-independent.

GD

β = 0.5

Significant at moderate retention — early integration uses a 2–3 word memory window.

GPT

No Effect

Regression-inclusive reading is driven by syntactic complexity, not memory decay.

TRT

β = 0.1

Strong local forgetting — re-reading is sensitive to recent context only.

reg

β = 0.7

Strongest effect (ΔAIC = 54) — regressions reflect medium-range memory integration failures.

Novelty & Contributions

First ET Validation of Lossy-Context Surprisal

Futrell et al. (2020) validated only on self-paced reading. We show it predicts eye-tracking across 5 distinct measures on ZuCo-1.

A Memory Gradient Across Reading Stages

Different measures have different optimal β — not just early vs. late, but a spectrum: GD (0.5) → TRT (0.1) → Regression (0.7).

GPT Dissociates from TRT

Go-past time is predicted by dependency length, not memory decay — showing regression-inclusive reading has distinct cognitive drivers.

Robust to Syntactic Confounds

Lossy surprisal effects survive after controlling for dependency length — capturing variance beyond structural complexity.

Remaining Work for Final Submission

Planned Analysis

- Increase lossy samples to 100+ for tighter β estimates
- Add random slopes to models where convergence allows
- Sensitivity analyses: vary fixation exclusion thresholds
- Correlation matrix of all predictors (check collinearity)
- Per-subject β optima — do individual readers differ?

Paper Writing

- Introduction: lossy memory & eye-tracking gap
- Methods: pipeline, ZuCo-1, GPT-2, glmmTMB spec
- Results: 5-measure β decay curves + tables
- Discussion: memory gradient interpretation
- Related work: GECO, UCL corpus, Futrell et al.

SUMMARY

Memory decay shapes reading differently across processing stages.

RQ1 ✓	Lossy surprisal explains ET variance beyond standard surprisal + dependency length
RQ2 ✓	Distinct optimal β for GD, TRT, regression — a memory gradient across measures
Novel:	First eye-tracking validation · GPT dissociates from TRT · 5-measure analysis